

ZS-M46

Koenigs Linearization and Half-Sided Modular Inclusions: The Cover-Level Translation, a Positive-Energy Standard Pair, and a Cocycle Realization on the Z-Spin Loxodromic Seam

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Theme / Code: Math Spine / ZS-M46 v1.5 (closure) · Downstream of ZS-A17 / ZS-F31 / ZS-F32 / ZS-A24 / ZS-F33 · Upstream of ZS-M47 / ZS-F37 / ZS-A32 · Program theorem 1 of 3 for the ZS-A31 §6.3 modular-depth line

Verification: exact/numerical checks **23/23 PASS** [SP 5 | K 12 | TW 3 | KLT 3]; imported-theorem / character / model / guard consistency checks **31/31 pass**. The suite does **not** machine-certify KH1–KH4, factor type, Haag duality, or cocycle realization — these are imported theorems or admissibility conditions. **Zero fitted parameters.** Central result: on the canonical Abel cover $u = \text{Log } \chi / \text{Log } \lambda$ the Z-Spin seam is the unit translation $u \mapsto u+1$ (PROVEN, Theorem M46.3A); the modular dilation is its positive-energy completion, not the seam itself. The intertwiner W_χ is a Hardy pullback (DERIVED-CONDITIONAL on admissibility KH1–KH4); the Fock HSMI (Theorem A) is DERIVED-CONDITIONAL on KH1–KH4; the realization on the ZS-A24 core is a Cocycle Realization Theorem (CRT-1 factor isomorphism, CRT-2 cocycle covariance, CRT-3 core lift, all DERIVED-CONDITIONAL on the injectivity of the parent factor; CRT-4 exact clock equality OPEN); the relative commutant $N' \cap M \cong A(0,1)$ is DERIVED-CONDITIONAL (model-fixed); $\mu/2\pi$ is the elliptic modulus height $h_K = \text{Im } \tau_K$, not a modular time. $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED*.

§0. Abstract

ZS-A17 named the one object the corpus lacked — an actual half-sided modular inclusion (HSMI) $N \subset M$ with a Takesaki resolution — and ZS-A31 §6.3 pre-registered its construction as ZS-M46. This paper supplies the kinematic construction and identifies precisely the single physical statement that remains open. (Earlier internal versions treated the radial contraction e^{-m} as the modular flow and $t_0 = \mu/2\pi$ as a “modular time”; v1.4 retracts that reading — see Version History — and identifies the canonical Abel lift as the unit translation.)

The role reversal (PROVEN). In the multiplicative Koenigs coordinate $\zeta = \chi(z)$ the Z-Spin iteration is the loxodromic map $\zeta \mapsto \lambda\zeta$ (two fixed points, abelian centralizer, no parabolic). On the canonical Abel cover $u = \text{Log } \chi(z) / \text{Log } \lambda$ (principal lift, $\log \lambda = -\mu + i\theta$, fixed by $\theta \in (0, \pi)$) the same seam is the unit translation $u \circ f = u + 1$ (Theorem M46.3A). No contradiction: Log is a universal covering, not a Möbius automorphism, so fixed-point counts need not match. The corpus datum is therefore the translation* $U(1)$ — the generator of the $(ax+b)$ group — and the modular dilation Δ^{it} is recovered afterward as its positive-energy completion (Borchers). The seam’s rotation θ enters a helical normal form: with $x = -\text{Re } w/\mu$ and $\eta = \text{Im } w + (\theta/\mu)\text{Re } w$ ($w = \text{Log } \chi$), one seam step is exactly $x \mapsto x+1, \eta \mapsto \eta$.

What is new vs. what is imported. The single novel move is the chain *Koenigs germ* $\rightarrow$ *canonical Abel translation* $\rightarrow$ *positive-energy standard-pair completion* — the identification of the corpus’s own i-tetration dynamics as the translation datum of a standard pair. Everything downstream is the application of established theorems: the Borchers 2π law; the standard-pair / HSMI correspondence (Longo;

Rieffel–van Daele); the Connes–Haagerup uniqueness of the injective Type III₁ factor; the Connes cocycle derivative; and the interval relative commutant (Hislop–Longo; Guido–Longo–Wiesbrock).

Status. W_χ is a Hardy pullback, so the positive-energy standard pair and the Fock HSMI (Theorem A) are **DERIVED-CONDITIONAL** on four admissibility conditions KH1–KH4 (bundled as a Suspension-Admissibility hypothesis). The realization on the ZS-A24 continuous core is a **Cocycle Realization Theorem**: factor isomorphism (CRT-1), modular-cocycle covariance (CRT-2), and core lift (CRT-3) are DERIVED-CONDITIONAL on the injectivity of the parent factor; only the *exact* clock equality (CRT-4) is OPEN, and it splits into a central-cocycle form (CRT-4a) and the stronger weight-preserving form (CRT-4b). The relative commutant $N' \cap M \cong A(0,1)$ is DERIVED-CONDITIONAL in a strongly-additive net. The absence of *every* normal conditional expectation, the reformulated clock gate C_{int} , the depth selector $t^* = Q$, and the metric-dimension-3 Dirac operator are explicit successor problems (NON-CLAIM / OPEN). This is the closure version of ZS-M46. Zero fitted parameters; $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED.

Epistemic Status Legend

Tag	Meaning
PROVEN	Explicit proof or exact verification; no undischarged assumption.
PROVEN-LOCAL	Proven on a neighbourhood of the attracting fixed point (a germ).
IMPORTED-PROVEN	Proven in the external literature and used without re-proof; cited.
DERIVED	Follows from PROVEN / IMPORTED-PROVEN results by stated steps; no new parameter.
DERIVED-CONDITIONAL	Derived modulo explicitly named, falsifiable conditions (e.g. KH1–KH4, parent injectivity, model choice).
OBSTRUCTION	A proven negative result: a candidate object fails a required property.
OPEN	A genuine, well-posed gap; may carry a finite-decision reduction.
NON-CLAIM	Explicitly outside this paper’s scope (deferred to a named successor).
CONSISTENCY-CHECK	A numerical, character, model, or scope check that does NOT by itself certify a theorem.
RETRACTED	A prior-version claim withdrawn in this version, with reason.

§1. Introduction

Three corpus lines converge on one object. ZS-A17 v1.5 proved the 3-metric is not reconstructible from spin or information and named the missing input as “a genuine one-sided (half-sided modular) inclusion.” ZS-F36 v2.1 left the metric-scale gate open with the closing path a modular depth $e^{-2\pi t^*}$. ZS-A31 v1.5 §6.3 pre-registered the program as $ZS-M46 \rightarrow ZS-M47 \rightarrow ZS-F37$. ZS-M46 supplies the first node: the construction of a half-sided modular inclusion from the Z-Spin loxodromic seam.

The construction rests on a single observation about the corpus’s own i-tetration dynamics, and on the disciplined use of established operator-algebra theorems for everything else. The observation is that the seam, correctly lifted, is a *unit translation* — exactly the generator a Borchers–Wiesbrock standard pair is built from. The paper proves this (§3), completes it to a positive-energy standard pair and a Fock HSMI conditionally on four admissibility conditions (§4–§6), realizes it on the ZS-A24 continuous core up to Connes-cocycle conjugacy (§6), and states plainly the one physical identification that remains open.

Locked data (ZS-M1, ZS-F31 App. D): $A = 35/437$, $Q = 11$, $(\dim Z, \dim X, \dim Y) = (2, 3, 6)$; $\kappa^2 = A/Q = 35/4807$; $z = 0.43828 + 0.36059 i$; $\lambda = f(z) = (i\pi/2)z$, $|\lambda| = 0.89151 < 1$, $\mu = -\ln|\lambda| = 0.1148346250$, $\theta = \arg \lambda^* = 2.2592495540$.

§2. The Koenigs germ (T1) and the local flow

Theorem T1 (Koenigs germ). [PROVEN-LOCAL]. $f(z) = \exp(i\pi z/2)$ has an attracting fixed point z with multiplier $\lambda = (i\pi/2)z$, $|\lambda| = 0.89151 < 1$ (ZS-M1). On a neighbourhood of z the Koenigs function χ ($\chi(z) = 0$, $\chi'(z) = 1$) solves $\chi \circ f = \lambda\chi$, and in $w = \text{Log } \chi(z)$ one iterate is $w \mapsto w + \log \lambda \pmod{2\pi i}$, $\log \lambda = -\mu + i\theta$, $\mu = 0.1148346250 > 0$, $\theta = 2.2592495540 \in (0, \pi)$.

Proof. Take logs of the Schröder equation; $\mu = -\ln|\lambda| > 0$ as $|\lambda| < 1$, and $\theta = \arg \lambda \in (0, \pi)$. $\square$

(Exact/numerical: K1–K12.) The discrete iterate embeds in the local holomorphic flow $f_t = \chi^{-1}(\lambda^t \chi)$ on the basin; a global $\mathbb{R}$ -flow on the original z -plane (global injectivity of χ , non-escape at negative time) is **OPEN** and is not needed — the operator-algebra construction uses the cover translation of §3, not the z -plane map.

§3. The obstruction and the role reversal

Obstruction Lemma. [PROVEN, negative]. The affine half-plane $K = \{w : \text{Re } w \leq c\}$ is not a standard real subspace: for $c = 0$ it is not closed under real scalar multiplication ($v = -1 \in K$ but $(-1)v = +1 \notin K$), and $K \cap iK = \{\text{Re} \leq 0\} \cap \{\text{Im} \leq 0\} =$ the third quadrant $\neq \{0\}$, violating $H \cap iH = \{0\}$; for $c \neq 0$ it is affine and omits the origin. (Consistency: SP6.) Hence a geometric half-plane cannot be second-quantized directly; the standard subspace must be built on a function space. §4 does so — after §3.1 identifies the correct generator.

§3.1 Sphere-level obstruction and cover-level translation (PROVEN): the role reversal

Theorem M46.3A. [PROVEN]. (a) *Sphere level.* In the multiplicative Koenigs coordinate $\zeta = \chi(z)$ the seam $\zeta \mapsto \lambda\zeta$ is loxodromic with two fixed points $\{0, \infty\}$; its Möbius centralizer is abelian and contains no parabolic subgroup. (b) *Cover level.* On the canonical Abel cover, with $u := \text{Log } \chi(z) / \text{Log } \lambda$ (principal lift, $\log \lambda^* = -\mu + i\theta$, $\theta \in (0, \pi)$), the seam is the unit translation

$$u \circ f = u + 1.$$

Proof. (a) A Möbius map is loxodromic iff it has two distinct fixed points; $\zeta \mapsto \lambda\zeta$ fixes $\{0, \infty\}$, and the centralizer of a loxodromic element in $\text{PSL}(2, \mathbb{C})$ is the abelian torus fixing the same pair, which has no parabolic element. (b) On the universal cover of the punctured basin, $\text{Log } \chi(f(z)) = \text{Log } \lambda + \text{Log } \chi(z)$, so $u \circ f = (\text{Log } \lambda + \text{Log } \chi) / \text{Log } \lambda = u + 1$. $\square$ (Exact/numerical: KLT1, residual 6×10^{-15} at the finite-iterate Koenigs limit; the identity itself is exact.)

No contradiction with the fixed-point count. $\zeta \mapsto \text{Log } \zeta$ is a universal covering, not a Möbius automorphism of the Riemann sphere; fixed-point number is a Möbius invariant, not a covering invariant. The two fixed points of $\zeta \mapsto \lambda\zeta$ and the single fixed point (∞) of $u \mapsto u+1$ coexist without conflict. (Consistency: KLT4.)*

The role reversal. The corpus datum is therefore the **translation** $U(1)$, the generator of the $(ax+b)$ group — not a dilation with a translation missing. The modular dilation Δ^{it} is recovered afterward as the positive-energy completion of this translation (§4). This is the central result of ZS-M46 and its one externally-novel move: *Koenigs germ* $\rightarrow$ *canonical Abel translation* $\rightarrow$ *positive-energy standard-pair*

completion. The earlier “missing-generator” reading survives only as clause (a), a scope-limited sphere-level statement.

§4. The positive-energy standard pair and the elliptic height h_K

The carrier and the Hardy pullback. Let $\mathcal{H}_u = H^2(\mathbb{C}_+)$ be the Hardy space of the cover strip and set $\mathcal{H}_K := \{F \circ u : F \in H^2(\mathbb{C}_+)\}$ with $\|F \circ u\| := \|F\|$. Then $W_\chi : \mathcal{H}_K \rightarrow H^2(\mathbb{C}_+)$, $W_\chi(F \circ u) = F$, and the seam composition C_f satisfies $W_\chi C_f W_\chi^{-1} = T(1)$, because $C_f(F \circ u) = F \circ (u+1) = (F \circ \tau_1) \circ u$. Via Paley–Wiener, $H^2(\mathbb{C}_+) \cong L^2(\mathbb{R}_+, dp)$ with $T(a)\psi(p) = e^{iap}\psi(p)$ and generator $P = \text{mult-by-}p \geq 0$. On the common invariant domain $C_{c\infty}(\mathbb{R}_+)$ the dilation $D = \frac{1}{2} + p\partial_p$ satisfies, by direct computation, $[D, P]\psi = (\frac{1}{2} + p\partial_p)(p\psi) - p(\frac{1}{2} + p\partial_p)\psi = p\psi = P\psi$, and $D^\dagger = -D$, giving the Borchers relation $\Delta^{it}U(a)\Delta^{-it} = U(e^{-2pt}a)$ exactly ($\Delta^{it} := e^{-2ptD}$). The verification’s monomial calculation is a *formal coefficient identity* for $[D, P] = P$; the operator-domain statement is analytic, not machine-certified. (*Exact/numerical: SP1–SP4, SP6; consistency SP5, HD1–HD2.*)

Theorem A (positive-energy standard pair). [split].

- (i) *Canonical Hardy standard pair exists. [IMPORTED-PROVEN].* The positive-energy $(ax+b)$ representation on $L^2(\mathbb{R}_+, dp)$ carries a unique irreducible standard pair (H, U) with $H \subseteq \mathcal{H}$ standard ($H + iH$ dense, $H \cap iH = \{0\}$) and modular operator Δ above (Rieffel–van Daele 1977; Longo 2008).
- (ii) *The Koenigs cover realizes that pair. [DERIVED-CONDITIONAL on KH1–KH4].* Under the Suspension-Admissibility conditions — (KH1) the lifted Koenigs domain is translation-invariant; (KH2) the representation $F \circ u$ is unique; (KH3) the seam reflection descends to an antiunitary involution J ; (KH4) the pullback norm is non-degenerate on the dynamical domain — W_χ is a unitary intertwiner and $(\mathcal{H}_K, U)$ is unitarily equivalent to the canonical pair.

Remark (why conditional). The standard-pair/HSMI machinery is imported and unconditional; the bridge that applies it to the Koenigs seam is exactly KH1–KH4, which are not proved here. Bundling them into a single **Suspension-Admissibility Lemma** — “the Abel lift admits a translation-invariant suspension domain on which the pullback into $H^2(\mathbb{C}_+)$ is faithful, norm-nondegenerate, and reflection-compatible” — and proving it would upgrade Theorem A(ii) to DERIVED. It is left as the paper’s principal admissibility hypothesis.

The elliptic height (retraction of ‘modular time’). The radial contraction factor e^{-t} is the contraction in the *multiplicative* Koenigs coordinate; **it is not identified with the Tomita modular flow**. Consequently the quantity $\mu/2\pi$ is not a “modular time of one iterate” (a reading used in earlier internal versions and here **RETRACTED**); it is the **elliptic modulus height**

$$h_K := \text{Im } \tau_K = \mu / (2\pi) = 0.0182765, \quad \tau_K = (\log \lambda^*) / (2\pi i) = \theta/2\pi + i\mu/2\pi,$$

the imaginary part of the $E_{\{\lambda\}}$ *modulus* (ZS-F33). The identity $e^{-2ph_K} = |\lambda|$ holds exactly but is a diagnostic relation among modulus data, not a statement about a modular clock. (*Consistency: SP5, TW1–TW3.*)

§5. The helical normal form (θ)

Helical normal form. [DERIVED]. In the log-Koenigs coordinate $w = \text{Log } \chi$ one seam step is $w \mapsto w - \mu + i\theta$. Setting $x = -\text{Re } w/\mu$ and $\eta = \text{Im } w + (\theta/\mu)\text{Re } w$ gives exactly

$$x \mapsto x + 1, \quad \eta \mapsto \eta, \quad \text{slope } \theta/\mu = 19.674.$$

So the seam is a **real unit translation** (x , the generator of §3.1–§4) carrying a **helical internal holonomy** (η , invariant, of slope θ/μ). The rotation θ is thus not a component of the modular generator but the internal cycle of the E_{λ} torus — $\text{Re } \tau_K = \theta/2\pi$ is its twist. A purely dilational ($\theta = 0$) germ would lose this internal cycle; the seam is loxodromic because the corpus carries both a translation and an internal holonomy. (Exact/numerical: KLT2, KLT3; consistency KLT4.)*

§6. The half-sided inclusion and the Cocycle Realization Theorem

§6.1 The Fock HSMI (Theorem A')

Theorem A' (Fock HSMI). [DERIVED-CONDITIONAL on KH1–KH4]. Given the standard pair (H, U) of §4, its second quantization (Longo 2008) yields a von Neumann algebra $M = \mathcal{R}(H)$ on Fock space with cyclic-separating vacuum Ω , and $N := U(1)MU(-1) \subset M$ is a standard HSMI with the Borchers relations. The construction is imported once (H, U) is in hand; its Z-Spin content is the realization of (H, U) by the Koenigs cover (Theorem A(ii)), hence the conditionality on KH1–KH4. It concerns the lifted/suspension dynamics; the original z -plane global flow is not claimed.

§6.2 The 2π law (T5) and the chain (T6)

T5 (the 2π law). [IMPORTED-PROVEN; instantiation DERIVED-CONDITIONAL]. The relation $\Delta^{it}U(a)\Delta^{-it} = U(e^{-2\pi t}a)$, with 2π fixed by Tomita–Takesaki, is imported (Borchers 1992; Wiesbrock 1993) and holds exactly in the §4 model. Its instantiation on the Z-Spin seam is DERIVED-CONDITIONAL on KH1–KH4 (via Theorem A'). ZS-M46 contributes not the 2π but the identification of the seam as the translation the law acts on.

T6 (the chain). [unit PROVEN; chain DERIVED-CONDITIONAL on KH1–KH4]. The identification $f \leftrightarrow U(1)$ is now a theorem (Theorem M46.3A: $u \circ f = u+1$ fixes the unit step *canonically*, not by choice), so the Wiesbrock relations generate the integer chain $N_k = U(k)MU(-k)$ with an intrinsic unit. The chain's existence as von Neumann inclusions is conditional on Theorem A' (hence KH1–KH4); the affine 2×2 check (B7) is a consistency shadow, not a von Neumann check. Selection of a terminal depth $k = Q$ is **NON-CLAIM** ($\rightarrow$ ZS-F37).

§6.3 Genuineness (T4)

T4 (genuineness). [split]. If Theorem A' holds, the modular group moves N ($\sigma_t(N) = N_{\{\exp(-2\pi t)\}} \neq N$), so by Takesaki (1972) **no Ω - or chosen-weight-preserving normal conditional expectation $M \rightarrow N$ exists** (DERIVED-CONDITIONAL). The absence of *every* normal conditional expectation (any state) needs an independent index / operator-valued-weight obstruction and is **OPEN**. “Genuine = no CE” is this paper's strengthened definition, flagged as such; the role separation from the ZS-F32 averaging expectations $E_{\text{face}} / E_{\text{reg}}$ (which are CE) stands.

§6.4 The relative commutant (Theorem C)

Theorem C (relative commutant). [DERIVED-CONDITIONAL, model-fixed]. Fix the realization to a strongly-additive, Haag-dual chiral net (e.g. the $U(1)$ -current net attached to the standard pair by GLW). Then $M = A(0, \infty)$, $N = A(1, \infty)$, and by Haag duality plus the intersection property $N' \cap M = A((0, \infty) \cap (-\infty, 1)) = A(0, 1)$ — a non-trivial Type III₁ factor with Ω cyclic-separating, whose interval modular flow (Hislop–Longo 1982) fixes $\{0, 1\}$ with endpoint rate $\pm 2\pi$. Strong additivity is what keeps the intersection non-trivial (a general HSMI can have trivial relative commutant), so the conclusion depends on both the model choice and, upstream, KH1–KH4. (Consistency: RC1–RC3.) This is the first

explicit operator-algebraic datum *selected for* the Z-Spin realization; $A(0,1)$ itself is a known structure of the chosen net, and ZS-M46's contribution is to attach it to the Koenigs lift.

§6.5 The Cocycle Realization Theorem

The realization on the ZS-A24 continuous core decomposes into four steps, replacing the over-strong “exact flow equality” of earlier drafts by the canonical cocycle relation. Let M_K be the second-quantized Koenigs factor and M_A the ZS-A24 parent Type III_1 factor $\mathcal{A}^{III}$.

CRT-1 (factor isomorphism). [DERIVED-CONDITIONAL]. M_K is a hyperfinite Type III_1 factor (Araki–Woods / Wiesbrock). If M_A is injective Type III_1 , then by Connes–Haagerup uniqueness there is a $*$ -isomorphism $\Phi : M_K \rightarrow M_A$. *Condition:* the injectivity/hyperfiniteness of M_A . (CLPW's stated output is a Type II_1 observer algebra; that the underlying de Sitter parent is injective Type III_1 is standard for free/generalized-free fields but is carried here as a condition, not a quoted conclusion.)

CRT-2 (cocycle covariance). [DERIVED-CONDITIONAL on CRT-1]. For weights φ_K, φ_A the Connes cocycle $u_t = (D(\varphi_K \circ \Phi^{-1}) : D\varphi_A)_t$ satisfies $\Phi \sigma_t^{\wedge\{\varphi_K\}} \Phi^{-1} = \text{Ad}(u_t) \circ \sigma_t^{\wedge\{\varphi_A\}}$: the two modular flows are cocycle-conjugate (Connes' Radon–Nikodym theorem). Defining u_t requires Φ , hence the dependence on CRT-1.

CRT-3 (core lift). [DERIVED-CONDITIONAL on CRT-1]. Crossed products by cocycle-conjugate actions are isomorphic, so Φ lifts to $M_K \rtimes_{\sigma} \mathbb{R} \cong M_A \rtimes_{\sigma} \mathbb{R} = C_{\omega}$, the ZS-A24 core.

CRT-4 (exact clock matching). [OPEN, two forms]. Two modular flows coincide *exactly* iff $(D\psi : D\varphi)_t \in Z(M)$; in a factor this is $(D\psi : D\varphi)_t = e^{it\cdot 1}$. Hence: **CRT-4a** (central-cocycle clock equality, $\sigma_t^{\wedge\{\varphi_K\}} \circ \Phi = \Phi \circ \sigma_t^{\wedge\{\varphi_A\}}$ up to a central phase) — OPEN; **CRT-4b** (normalized weight-preserving equality, $u_t = 1$) — OPEN, strictly stronger. (*Consistency: CO1–CO4.*)

Theorem B, split. [B_abs DERIVED-CONDITIONAL; B_phys OPEN]. *Abstract transport* (B_{abs}): granting CRT-1 and KH1–KH4, Φ carries the Fock HSMI $N_K \subset M_K$ to a standard HSMI $\Phi(N_K) \subset M_A$ on the core — DERIVED-CONDITIONAL. *Physical identification* (B_{phys}): that $\Phi(N_K)$ equals the algebra/weight ZS-A24 physically designates for the seam ($\Phi \sigma_t^{\wedge K} \Phi^{-1} = \sigma_t^{\wedge\{A24\}}$ on the nose) requires CRT-4 — OPEN. So the inclusion *structure* transports (up to cocycle conjugacy), while the *physical clock identification* is the single remaining physical claim of ZS-M46.

§7. The register measure (T7a) and the reformulated clock gate (T7b)

T7a (register measure). [A24-derived conditional corollary]. On the ZS-A24 Type II_1 corner $M_{\text{obs}} = eC_{\omega}$ the normal tracial state τ is unique (Murray–von Neumann); under the register embedding ι the eleven rank-one microstate projections carry $\tau(e_{\alpha}) = 1/11$, so the induced density is $\rho_Q = I_Q/Q$. This follows from the A24 matrix embedding and the II_1 unique trace, **not** from the HSMI; it populates the ZS-M44 route (ii) only as the unique KMS state of the *trivial* (tracial) flow. A concrete route to promote it to DERIVED is to verify, on the actual ZS-F31 GKLS generator L , that $L(I_Q/Q) = 0$ with $\ker L = \mathbb{C} \cdot (I_Q/Q)$ (primitivity) — a finite computation deferred to ZS-M44/F31. (*Consistency: C4.*)

T7b (clock gate C_{int}). [OPEN-REFORMULATION]. The pre-role-reversal ratio test $\mu/\omega_Z = \theta/v_Z$ read the seam as a dilation and is **not used** after the role reversal. With the seam a translation, the clock gate must be rewritten in terms of the translation generator P_K , the A24 modular Hamiltonian K_A , and the central Connes cocycle — e.g. as $\Phi(P_K) = P_A$, or $(D\varphi_K \circ \Phi^{-1} : D\varphi_A)_t \in \mathbb{T} \cdot 1$ (a central phase). This coincides with CRT-4a. The residual is the explicit ZS-F31 seam diagonalization; C_{int} is **OPEN** as a finite, well-posed reformulated computation, not the old ratio comparison. (*Consistency: CII.*)

§8. The scale boundary (NON-CLAIM)

ZS-M46 provides only dimensionless modular structure. (N1) depth selection $t = Q \rightarrow ZS-F37$ (T6 gives at most a conditional ladder; no terminal rung). (N2) the metric-dimension-3 Dirac operator $\rightarrow ZS-M47$ (one inclusion gives 1D chiral data; ≥ 3 inclusions in modular position are needed for a 3-metric). (N3) the absolute scale $\rightarrow$ the ZS-F36.T7 / ZS-A27 / ZS-A28 Buckingham- π no-go is inviolate: the model fixes the shape (the modulus τ_K , the height h_K , the exponent 2π , the unit step) but never the size. The only dimensionful comparison, $e^{-2\pi\eta}$, is quarantined behind the regression firewall (C7) and feeds nothing. (Consistency: C7, C8.)*

§9. Zero-parameter and cross-version audit

Zero fitted parameters. $\mu = 0.1148346250$, $\theta = 2.2592495540$, $|\lambda^*| = 0.89151$ are ZS-M1; $h_K = \text{Im } \tau_K = \mu/2\pi$ and the slope θ/μ are definitions on those; $A = 35/437$, $Q = 11$, $\kappa^2 = 35/4807$ are LOCKED; 2π is the Tomita–Takesaki constant. The one observational number, $e^{-2\pi\eta}$, is firewalled (C7). No fudge factor.

Cross-version dependency trace. ZS-M1: used only via λ and the local germ; z and $|\lambda| < 1$ untouched, so the $M1 \rightarrow S1 \rightarrow U1$ chain is safe. ZS-F31: C_{int} is reformulated (T7b), not evaluated; no value altered. ZS-F32: F32.1 respected; E_{face}/E_{reg} preserved. ZS-F33: $E_{\{\lambda\}}$ periods and modulus τ_K used verbatim (§4–§5). ZS-A17: the inclusion-source input is realized up to Connes-cocycle conjugacy and the KH/CRT conditions — the metric NO-GO is untouched. ZS-A24: the II_∞ core, II_1 corner, and $\mathfrak{t}$ are used verbatim; the parent Type III_1 factor is the CRT-1 target. ZS-A31: the 2π is contextualized as imported; the metric-scale gate and t^* stay OPEN/NON-CLAIM. No upstream status is reversed.

Observational non-collision. ZS-M46 makes no dimensionful prediction, so it cannot collide with Planck 2018 Λ CDM or Standard-Model couplings; the frozen $w = -1$ branch (via ZS-F33/A28) is unchanged. DESI DR2 dark-energy preferences bear on the ZS-A31 scale line, not on the dimensionless structure here.

§10. Falsification gates

Tier 1 — mathematical / immediate rejection.

F-M46.1. If $\mu \leq 0$, the attracting Koenigs germ and the chosen one-sided positive-energy orientation fail; the algebraic cover identity $u \circ f = u+1$ itself need not fail (it holds whenever $\log \lambda \neq 0$). PASS:* $\mu = 0.1148 > 0$ (K9).

F-M46.2. If $u \circ f \neq u+1$ on the cover, the role reversal fails. PASS: KLT1 (residual 6×10^{-15} , germ-approx limit; identity exact).

F-M46.3. If the affine half-plane were a standard subspace, the Obstruction Lemma is false. Result: it is not (SP6); the direct half-plane route is excluded.

Tier 2 — admissibility / structural.

F-M46.4. If no translation-invariant suspension domain exists (KH1–KH4 unsatisfiable), W_χ is not a standard-pair intertwiner and Theorem A(ii), A' collapse to the canonical pair only. Status: OPEN admissibility — the paper's principal conditional.

F-M46.5. If M_A is not injective Type III_1 , CRT-1 fails and the core realization is void. Status: OPEN condition on the parent factor.

F-M46.6. If $(D\phi_K \circ \Phi^{-1} : D\phi_A)_t \notin \mathbb{T} \cdot 1$ for every admissible Φ , then no clock identification exists (CRT-4a fails) and B_{phys} is refuted. Status: OPEN — the single remaining physical gate.

Tier 3 — scope / numerology guard.

F-M46.7. Any derivation-module equation setting $k = Q$, or fixing a dimensionful scale from A , Q , topology alone, is void by Buckingham- π . *PASS*: C8; e^{-2p_0} firewalled (C7).

F-M46.8. If consistency checks (character identities, imported-theorem echoes, guards) were counted as theorem certification, the verification would be over-stated. *PASS*: they are reported as a separate tier (31 consistency vs 23 exact/numerical).

§11. Conclusion

ZS-M46 closes with a clean separation of what is new from what is imported. **New**: the single chain *Koenigs germ* $\rightarrow$ *canonical Abel translation* $\rightarrow$ *positive-energy standard-pair completion* — the identification of the corpus's i-tetration seam as a unit translation on the Abel cover (Theorem M46.3A, PROVEN), from which the positive-energy standard pair, the Fock HSMI, and the helical normal form follow. **Imported**: the Borchers 2π law, the standard-pair/HSMI correspondence, the Connes–Haagerup uniqueness of the injective Type III₁ factor, the Connes cocycle derivative, and the interval relative commutant. ZS-M46's contribution is the bridge, not the imported machinery.

The honest terminus is a conditional realization with one physical residual. The kinematic construction is complete modulo the Suspension-Admissibility conditions KH1–KH4; the core realization holds up to Connes-cocycle conjugacy (CRT-1–3, conditional on the parent factor's injectivity); and the modular dilation is recovered as the completion of the seam translation, with $\mu/2\pi$ read correctly as the elliptic height h_K , not a modular time. Four items are explicit successor problems and are **not** pursued further here: exact clock equality (CRT-4 / B_{phys}), the absence of every normal conditional expectation, the reformulated C_{int}, and the depth selector $t^* = Q$. This is the first of the three ZS-A31 §6.3 program theorems, closed to the extent its evidence supports and no further. $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED.

§12. Claim ledger (v1.4, final)

#	Claim	Status
T1	Koenigs germ: iterate = translation $w \mapsto w + \log \lambda^*, \mu > 0$	PROVEN-LOCAL
Obstruction	affine half-plane is not a standard subspace	PROVEN (negative)
M46.3A(a)	sphere-level: loxodromic, abelian centralizer, no parabolic	PROVEN (scope-limited)
M46.3A(b)	cover-level: $u \circ f = u + 1$ (unit translation)	PROVEN (role reversal)
—	positive-energy model $P \geq 0$, $[D, P] = P$, Borchers 2π	IMPORTED-PROVEN + verified
Helical	$\theta: x \mapsto x + 1, \eta \mapsto \eta$, slope θ/μ	DERIVED
h_K	$\mu/2\pi = \text{Im } \tau_K$ (elliptic height; 'modular time' retracted)	DERIVED (relabelled)
W_χ	Hardy pullback intertwiner	DERIVED-CONDITIONAL (KH1–KH4)
A(i)	canonical Hardy standard pair exists	IMPORTED-PROVEN
A(ii)/A'	Koenigs cover realizes it; Fock HSMI	DERIVED-CONDITIONAL (KH1–KH4)
T4	no chosen-weight CE / any CE	DERIVED-COND / OPEN
T5	$\Delta^{it}U(a)\Delta^{-it} = U(e^{-2p_t}a)$	IMPORTED-PROVEN; inst. DERIVED-COND
T6	unit-depth chain N_k (unit canonical)	unit PROVEN; chain DERIVED-COND
Thm C	$N' \cap M \cong A(0, 1)$, III ₁ , rate $\pm 2\pi$	DERIVED-CONDITIONAL (model-fixed)

#	Claim	Status
CRT-1	factor isomorphism $\Phi : M_K \cong M_A$	DERIVED-CONDITIONAL (injectivity)
CRT-2/3	cocycle covariance; core lift	DERIVED-CONDITIONAL on CRT-1
CRT-4a/4b	exact clock equality (central / weight-preserving)	OPEN / OPEN (stronger)
B_abs / B_phys	abstract transport / physical identification	DERIVED-COND / OPEN
T7a	$\rho_Q = I_Q/Q$	A24 corollary (GKLS-primitivity path)
T7b	C_{int}	OPEN-REFORMULATION
N1/N2/N3	$t^*=Q$ / Dirac / absolute scale	NON-CLAIM

§13. Verification

zs_m46_verify_v1_4.py (SymPy + mpmath 60-digit + NumPy), two honestly separated tiers. **The suite does not machine-certify KH1–KH4, factor type, Haag duality, strong additivity, or cocycle realization** — these are imported theorems or admissibility conditions.

EXACT / NUMERICAL — 23/23 PASS. *SP (5):* positivity $\min \sigma(P) > 0$; $[D,P]=P$ (a formal coefficient identity, common domain $C_{c\infty}(\mathbb{R}_+)$) and $D^\dagger = -D$; the Borchers relation; the affine half-plane obstruction (SP6). *K (12):* the Koenigs / ZS-M1 germ data. *TW (3):* the $E_{\{\lambda\}}$ modulus and factorization. *KLT (3):** the cover translation $u \circ f = u+1$ and the helical split $x \mapsto x+1$, $\eta \mapsto \eta$.

CONSISTENCY — 31/31 pass (do NOT certify the theorems). *SP5:* the diagnostic identity $e^{-2ph}_K = |\lambda|$ ($h_K = \text{Im } \tau_K$, not a modular time). *HD1–HD2:* the translation-character identity for W_χ (not KH1–KH4 / unitarity / surjectivity). *RC1–RC3:* imported Haag-duality / Hislop–Longo echoes for $A(0,1)$. *KLT4:* the sphere-vs-cover fixed-point statement. *CO1–CO3:* the cocycle chain identity, hyperfinite III_1 uniqueness, and crossed-product isomorphism; *CO4a/CO4b:* the two exact-equality gates (automorphism $u_t \in {}^T I$, and weight-preserving $u_t = 1$). *CI1, C6 (guards):* the C_{int} gate reformulated via CRT-4a, the old ratio test absent; *CI2:* the h_K note. *B (8), C (6), R (3):** affine 2×2 shadows, corpus arithmetic + firewall, cross-version identities.

§14. Acknowledgements & Code Availability

zs_m46_verify_v1_5.py reproduces all 54 checks (23 exact/numerical + 31 consistency). This work used AI tools (Anthropic Claude) for external-literature search, cross-paper integration, symbolic/numerical verification, and drafting, under Kenny Kang’s editorial direction. The role reversal (the cover-level translation), the CRT decomposition, and the honest verification re-classification were made in response to iterated external technical review; the author assumes full responsibility for all content.

§15. Appendix A — Deep-exploration (issue-tree) record

Step 0 (long list, 7). (1) the kinematic lift — what is the seam’s actual generator; (2) the positive-energy realization; (3) the operator-algebra/core realization; (4) downstream selection (chain, register, C_{int} , t^*); (5) the relative commutant; (6) genuineness / no-CE; (7) global flow. Dropped as standalone: (7) resolved by the cover lift (PROVEN on the cover; z-plane OPEN, not needed); (5),(6) folded into (3),(4).

Step 1 (MECE, 4). I1 kinematic lift (the Abel cover); I2 positive-energy standard pair (Hardy pullback); I3 algebra/core realization (CRT); I4 downstream (chain / register / C_{int} / t^*).

Steps 2–3 (tree + status). I1 $\rightarrow$ I2 $\rightarrow$ I3 $\rightarrow$ I4. I1: Theorem M46.3A PROVEN. I2: A(i) IMPORTED-PROVEN, A(ii)/A' DERIVED-CONDITIONAL on KH1–KH4. I3: CRT-1/2/3 DERIVED-CONDITIONAL, CRT-4 OPEN; Theorem C DERIVED-CONDITIONAL (model-fixed). I4: T5 imported+conditional, T6 unit PROVEN/chain conditional, T7a A24 corollary, T7b OPEN-reformulation, $t^*=Q$ NON-CLAIM.

Step 4 (convergence). Across versions the number of nodes whose status *changed* per re-traversal fell 9 $\rightarrow$ 4 $\rightarrow$ 3 $\rightarrow$ 0 (a convergence criterion on the *change count*, not on the residual OPEN count) as the role reversal (v1.3) and the status/verification corrections (v1.4–v1.5) were applied, with no reversals; the exploration **converges**. The residual OPENs (KH1–KH4, CRT-1 injectivity, CRT-4/B_phys, any-CE, C_int, $t^*=Q$) are genuine successor problems requiring new input or explicit finite computation, not decomposition failures.

Step 5 (value). Converged + corpus non-collision (the downgrades and the obstruction remove over-claims; A24/F31/F33 used verbatim) + anti-numerology (the new results — $u \circ f = u+1$, the helical split — are parameter-free; the imported machinery adds none). The value is an honest, externally-legible construction paper whose one novel connection is precisely delimited from its imported components.

Appendix B — Key numerics

Quantity	Value	Role
z^* (i-tetration fixed point)	$0.438282936727 + 0.360592471871 i$	ZS-M1 PROVEN
$\lambda = (i\pi/2)z$	$ \lambda^* = 0.891513565776 < 1$	multiplier, attracting
$\mu = -\ln \lambda^* $	0.114834624996	contraction; sets the translation unit; > 0
$\theta = \arg \lambda^*$	2.259249553900	internal holonomy; $\in (0, \pi)$
$\tau_K = \theta/2\pi + i\mu/2\pi$	$\text{Im } \tau_K = \mu/2\pi$	$E_{\{\lambda^*\}}$ elliptic modulus (F33)
$h_K = \text{Im } \tau_K = \mu/(2\pi)$	0.018276498210	elliptic modulus height (NOT a modular time)
slope θ/μ	19.673944	helical normal-form slope
$e^{-2\pi\theta}$ (firewalled)	9.632×10^{-31}	A31 §6.3 comparison only; no derivation use

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§17. Version History

v1.0 (July 2026): Initial internal draft (never publicly finalized). Read the radial contraction e^{-m} as a modular dilation, treated the affine half-plane as a standard datum, and claimed a genuine standard HSMI (DERIVED). Consolidated from internal Z-Spin Collaboration research notes.

v1.1: Retracted the standard-subspace step (Obstruction Lemma, PROVEN negative); built the explicit positive-energy model; introduced $t_0 = \mu/2\pi$ as a “modular time” (DERIVED-CONDITIONAL on an intertwiner W_χ , then OPEN).

v1.2: Added the Missing-Generator Lemma (the abelian seam supplies only a dilation), the relative commutant $N' \cap M \cong A(0,1)$, and the Central Realization Hypothesis consolidating three open items.

v1.3: The role reversal — on the Abel cover $u = \text{Log } \chi / \text{Log } \lambda^*$ the seam is the unit translation $u \circ f = u+1$ (PROVEN). W_χ became a Hardy pullback; the CRH was replaced by the Cocycle Realization Theorem; t_0 was reinterpreted as the elliptic height.

v1.4 (July 2026): Closure release. Purges the residual v1.0–v1.2 framing that conflicted with the role reversal, and corrects status over-claims flagged in review: $t_0 = \mu/2\pi$ is uniformly the elliptic height $h_K = \text{Im } \tau_K$, not a modular time (§4, Appendix B, verification SP5 moved to consistency); Theorem A is split into an IMPORTED-PROVEN canonical pair and a DERIVED-CONDITIONAL Koenigs realization (KH1–KH4, bundled as a Suspension-Admissibility hypothesis); Theorem B is split into abstract transport (DERIVED-CONDITIONAL) and physical identification (OPEN); CRT-2/3 are made DERIVED-CONDITIONAL on CRT-1, and CRT-4 is split into central-cocycle (4a) and weight-preserving (4b) forms, correcting the over-strong “iff $u_t = 1$ ”; Theorem C is DERIVED-CONDITIONAL (model-fixed); C_{int} (T7b) is marked OPEN-REFORMULATION in terms of P_K , K_A , and the central cocycle; and the verification is re-classified into 24 exact/numerical and 29 imported-theorem/character/model/guard consistency checks, with an explicit disclaimer that KH1–KH4,

factor type, Haag duality, and cocycle realization are not machine-certified. This closes ZS-M46; the residual problems (CRT-4/B_phys, any-state CE, C_int, $t^*=Q$, the metric-dimension-3 Dirac operator) are named successors. Zero fitted parameters; (A, Q, dim Z) = (35/437, 11, 2) LOCKED.

v1.5 (July 2026): Final closure release. Editorial/consistency pass making the paper text and the verification script exactly agree, with no new mathematics. Corrections: (i) the verification classification is fixed — SP5 (the h_K identity) is moved from exact to consistency, giving **23 exact/numerical [SP 5 | K 12 | TW 3 | KLT 3] + 31 consistency = 54**, matching the script; (ii) the retracted C_int ratio test is removed from the code (C6, C11 become guards registering C_int as OPEN-REFORMULATION via CRT-4a); (iii) the exact-clock gate is split in the code as well as the text into CO4a (automorphism equality, $u_t \in {}^T \cdot 1$) and CO4b (weight-preserving, $u_t = 1$); (iv) the common invariant domain is named $C_{\infty}(\mathbb{R}_+)$ and the monomial calculation is labelled a formal coefficient identity, not machine-certified; (v) the convergence statement is corrected to a *change-count* criterion ($9 \rightarrow 4 \rightarrow 3 \rightarrow 0$), not a residual-OPEN count; (vi) F-M46.1 is scope-limited ($u \circ f = u+1$ does not itself require $\mu > 0$); (vii) missing dependency references (Murray–von Neumann, ZS-F36, ZS-A27/A28, a Hardy/Paley–Wiener source) are restored; (viii) sections are renumbered (§16 References, §17 Version History) and the principal branch $\log \lambda^* = -\mu + i\theta$ is stated. This is the final closure of ZS-M46. Zero fitted parameters; (A, Q, dim Z) = (35/437, 11, 2) LOCKED.